

Development and Quality control of Novel Scintillating Tiles for Super MuSR

Ben Thompson
University of Bristol
Detector Systems Group
ISIS Muon and Neutron Spallation Source, STFC, UKRI

Abstract

This report investigates the development and quality control of novel scintillating tiles constructed for Super MuSR. Tiles are made from scintillating plastic, wavelength shifting fibre, and optically clear resin. Tiles of lengths ranging from 30mm to 210mm were investigated. Select tiles were verified on the ISIS beamline as being suitable for the final Super MuSR detector and subsequently set as benchmark tiles. Unverified tiles are tested against benchmark tiles using a pair of Photomultiplier tubes and a collimated ^{90}Sr source. For the 63mm and 105mm tiles, they performed on average 1.5% and 2.9% better than the benchmark tiles, respectively, while 210mm tiles performed on average 1.9% worse. The 30mm tiles performed 23.7% better than the selected benchmark tile, which was selected separately from the beamline experiments. Tiles that fell more than two standard deviations below the benchmark mean were separated for further investigation. 1.4%, 13.9%, 7.3% and 18.9% of the 30mm, 63mm, 105mm, and 210mm tiles were flagged for investigation, with 8.6% of all tested tiles being investigated. This report discerns that RAL71 resin is the most suitable choice for tiles with length $\leq 63\text{mm}$, and RT152 resin for tiles with length $> 63\text{mm}$. Results suggest that the two main modes of failure are poor optical coupling between the plastic scintillator and WLSF in the tiles and double handling/re-lapping of tiles. Both of these issues have been addressed, helping keeping Super MuSR on track.

1 Acknowledgements

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2 Introduction

The ISIS Neutron and Muon source is a world-class research facility which conducts 1200 experiments yearly in fields such as material science, life sciences, quantum materials and energy technologies. This generates 600 yearly publications [1] [2]. The Detector Systems Group (DSG) is the department dedicated to the maintenance and development of current and future detectors at ISIS [3], and is playing a key role in four new instruments being developed as part of ISIS's Endeavour programme. The endeavour program is the decade-long plan that ISIS aims to follow to drive research in three key areas: materials of the future, clean growth, and life sciences [4]. One of these instruments is Super MuSR, which is replacing the current MuSR instrument. Both of these instruments are named after μ SR, a technique that stands for Muon Spin Rotation, Relaxation, and Resonance. These are characteristic behaviours of the muon, and can be used to probe the magnetic composition of materials. This report aims to outline part of the research and development process of Super MuSR, with a focus on the development and quality control of novel scintillating tiles used to collect information from the positrons emitted by muon decay.

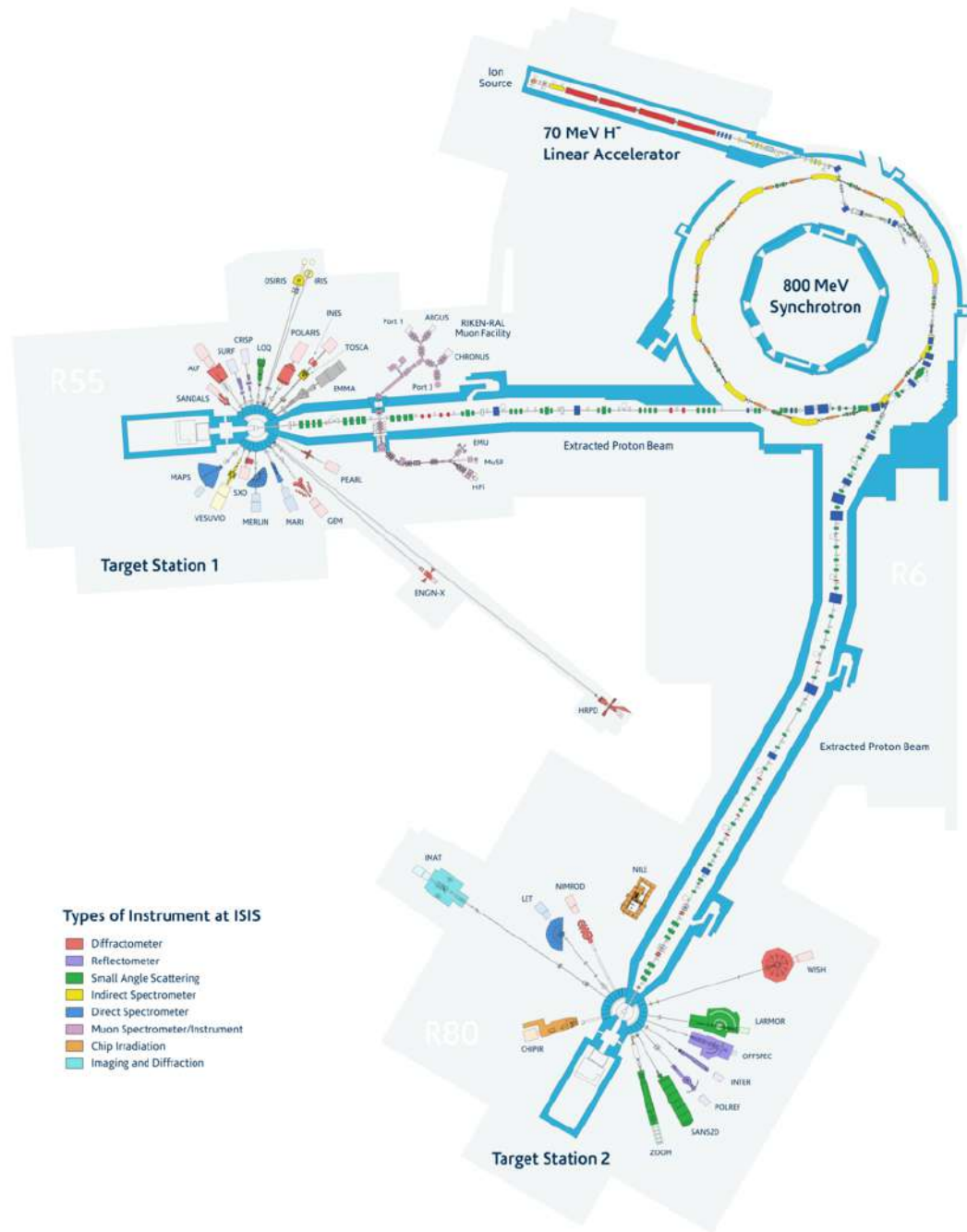


Figure 1: A schematic of the ISIS Neutron and Muon Source. Note the section in purple are the muon arms of ISIS, located in Target Station 1. MuSR is situated on the southern Muon arm [5].

3 Theory

3.1 Muon Production

MuSR and Super MuSR require a beam of positive, spin polarised muons (μ^+) to conduct μ SR experiments. H^- ions are accelerated firstly by a series of radio frequency quadrupoles, and then a 70MeV linear accelerator. Then they are then stripped of their electrons and fed into a 800MeV synchrotron which accelerates them to 84% the speed of light [5] [6]. The protons collide with a thin carbon target to produce positive pions (π^+) which decay into muons with the following decay path [7]:

$$p + p \rightarrow p + n + \pi^+ \quad (1)$$

$$\pi^+ \rightarrow \mu^+ + \nu_\mu \quad (2)$$

Crucially, muons must arrive highly spin-polarised at the sample, this is done by selecting "surface muons". This is where π^+ decay at rest close to the surface of the carbon target. The π^+ decays via the weak interaction and violates parity, so muons are emitted with their spin anti-parallel to their momentum [8]. A velocity selector then guides these polarised muons into the sample. [?] ?

3.2 μ SR Essentials

μ SR is a technique that utilises the magnetic spin of muons to probe materials. μ^+ have a spin of $1/2$, a mass of $105.658 \text{ MeV}/c^2$, and a mean lifetime $\tau_\mu = 2.197 \mu\text{s}$ [9]. The muons magnetic moment ($\boldsymbol{\mu}$) is directly proportional to the intrinsic angular momentum of the muon ($\boldsymbol{S}$):

$$\boldsymbol{\mu} = \gamma \boldsymbol{S}. \quad (3)$$

Where the constant of proportionality is the gyromagnetic ratio γ , for the muon:

$$\frac{\gamma_\mu}{2\pi} = 135.5 \text{ MHz T}^{-1}, \quad (4)$$

μ^+ come to a rest inside the sample through only electrostatic interactions, maintaining polarisation [10]. The muons behave like small bar magnets, and align their spin with the local magnetic field at their stopping site. The local magnetic field ($\boldsymbol{B}_{local}$) is the sum of applied and internal fields in the sample and exerts a torque on the magnetic moment on the muon, causing Larmor precession (see Figure 2a). The Larmor precession frequency (ω) is given by:

$$\omega = \gamma_\mu B_{local}. \quad (5)$$

ω is measured through the μ^+ decaying into a positron (e^+) [7]:

$$\mu^+ \rightarrow e^+ + \bar{\nu}_\mu + \nu_e. \quad (6)$$

This decay also violates parity, so positrons are emitted preferentially in the direction of the muon spin, according to the distribution in Figure 2b. Two banks of detectors, denoted as the forward and backward detector, surround the sample (example geometry seen in Figure 3a) to measure the time dependent asymmetry $A(t)$:

$$A(t) = \frac{N_F(t) - \alpha N_B(t)}{N_F(t) + \alpha N_B(t)} = a_0 P_z(t). \quad (7)$$

Where N_F and N_B are counts in the forward and backward detectors, α is a balancing constant to account for detector efficiencies, and $P_z(t)$ is the longitudinal polarisation function. The resultant plot for an ensemble of polarised muons is an exponentially decaying curve that oscillates as the muons precess in the sample (see Figure 3b). This allows for the calculation of the Larmor frequency and subsequently the local magnetic field strength can be extracted [7]. Positron events are typically detected by blocks of plastic scintillator, which produce light when excited by ionising radiation [11], this light is then collected and measured.

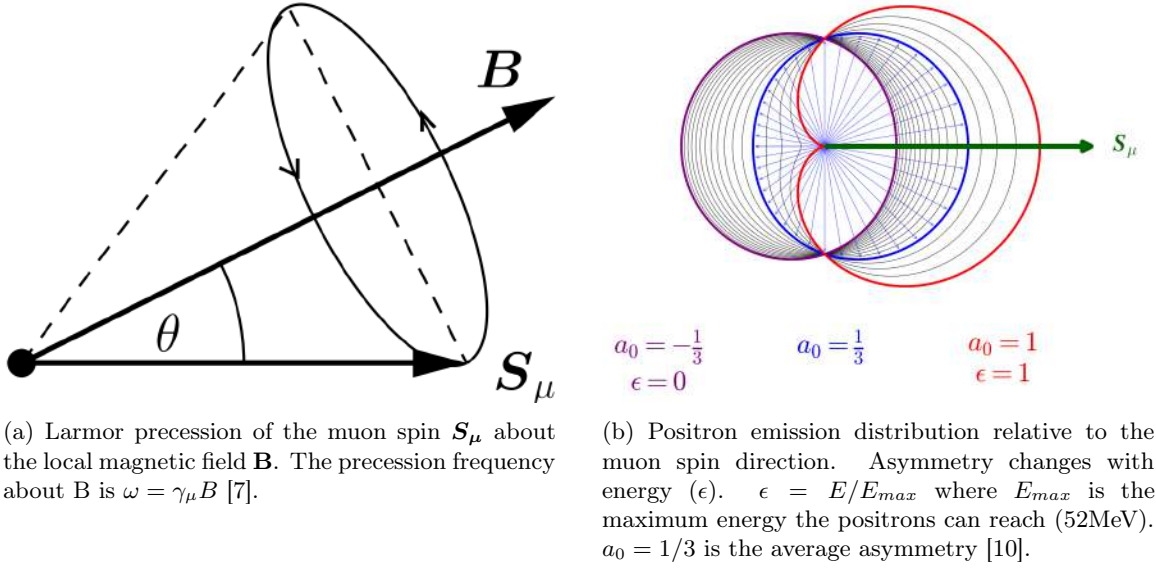


Figure 2: (a) Larmor precession of the muon spin in a local magnetic field, and (b) Asymmetric probability distribution of positron emission.

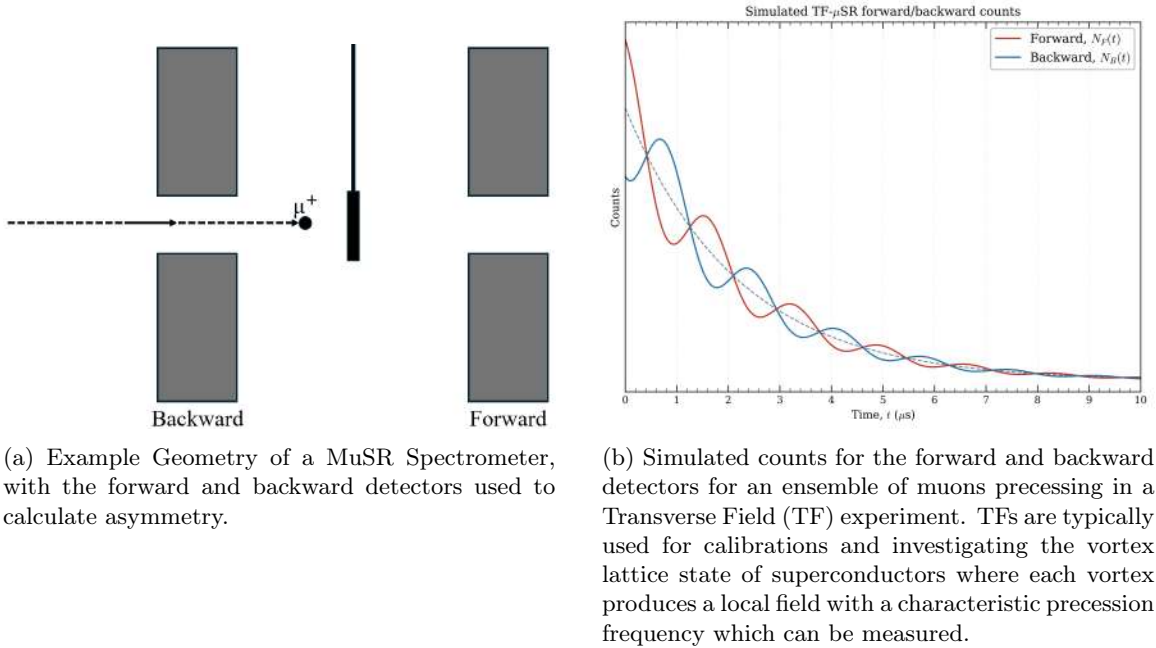


Figure 3: (a) Example detector geometry, and (b) Example counts for forward and backward detectors

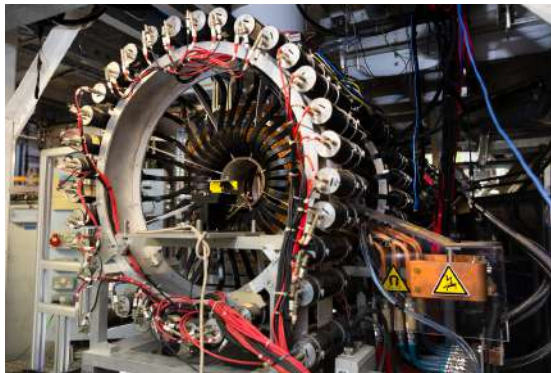
3.3 μ SR Experimental Geometries

The three main experimental geometries for a μ SR experiment are Zero Field (ZF), Longitudinal Field (LF), and Transverse Field (TF) experiments. ZF measurements utilise compensation magnets such that only the magnetic field of the sample itself remains, the muons depolarise over time known as muon relaxation, applications include study of magnetic and ion dynamics [12]. LF measurements apply a magnetic field in the direction of the initial muon spin, suppressing static magnetic fields and isolating fluctuating magnetic fields that withstand larger applied LFs. Varying LF strength reveals where static or dynamic order dominates, allowing characterisation of spin dynamics near magnetic phase transitions. TF measurements apply fields perpendicular to initial spin direction. TF measurements are used in instrument calibration (Figure 3b) and investigating the vortex lattice state of superconductors [13].

3.4 Detector Design

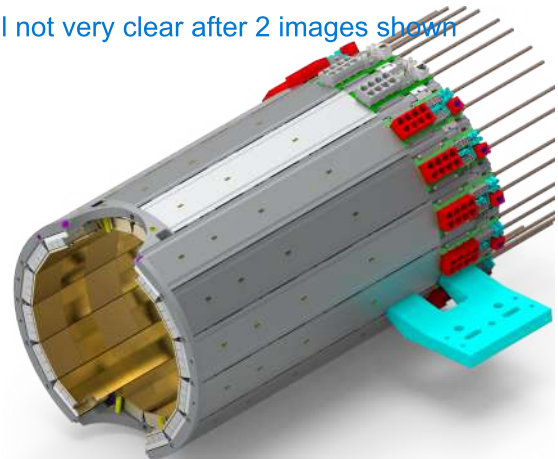
3.4.1 MuSR

MuSR is the current general purpose Muon Spectrometer at ISIS, it works by utilising two barrels of 32 detectors. Positrons are detected using a plastic scintillator, and perspex light guide coupled to a Photomultiplier tube (PMTs). The PMTs are clad in magnetic shielding and positioned on the outside of the instrument's electromagnets that can apply up to 2500G. PMTs convert light produced from positron events into an amplified current pulse via a photocathode and dynode chain [14] [15]. MuSR currently detects approximately 20% of the all emitted positrons.



(a) The Current MuSR instrument. Visible are the PMTs that are in a ring far outside of the sample area, with the light guides leading to inside the main magnet near the sample environment.

still not very clear after 2 images shown



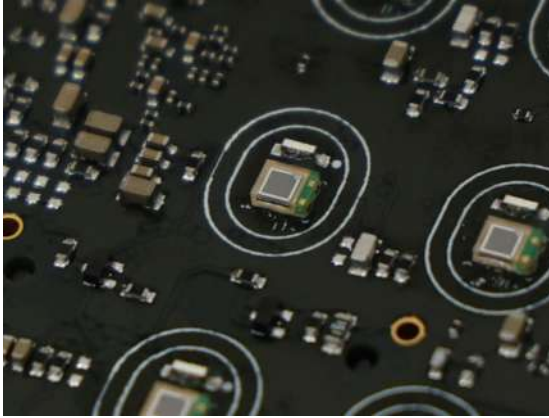
(b) A schematic for one barrel of the new Super MuSR detector, the Left Hand Side (LHS) contains the many detector elements. On the RHS there are the wiring and cooling pipes, some of which are hidden for clarity.

Figure 4: (a) Image of the old MuSR instrument (b) Schematic for its replacement, Super MuSR

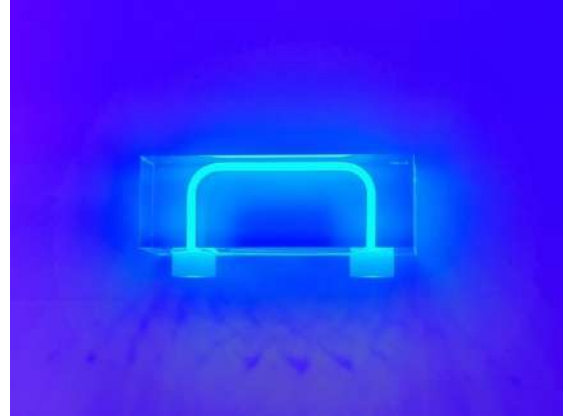
how one replaces another? ring replaces by a barrel?

3.4.2 Super MuSR

Super MuSR will replace the current MuSR instrument, improving count rate by a factor of 20, in addition to a 6-10 $\times$ increase in temporal resolution, facilitating high-statistics measurements to be taken on a shorter timescale. Time resolution improvements allow for a greater range of internal fields to be measured. This improves the range of materials Super MuSR can investigate, including groups



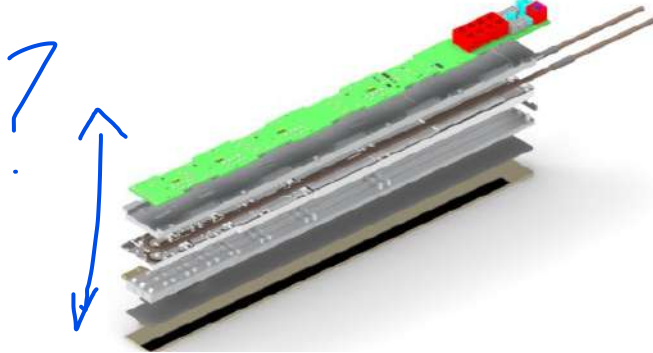
(a) A close up image of a SiPM from one of the Super MuSR detector Modules.



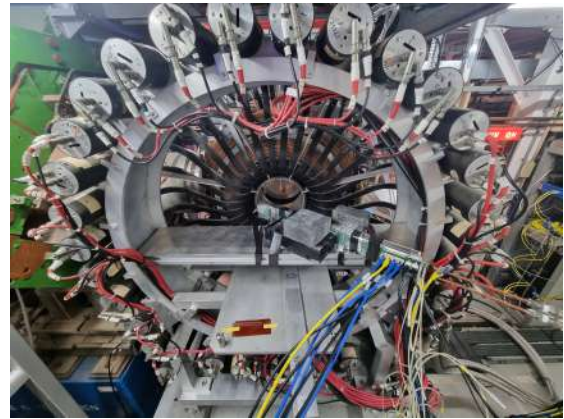
(b) An unwrapped 30mm scintillating tile.

Figure 5: (a) Image of a SiPM (b) Image of an unwrapped scintillating tile.

such as oxides, as well as greater scope in superconductor vortex lattice studies. The entire detector is being moved inside of the current MuSR main magnet which is being upgraded to facilitate greater TF and LF field strengths. Better data collection rates allow for further scope in battery material research, and improved use of RF measurements (where radio wave pulses are applied to the sample between muon arrival, allowing the determination of hyperfine coupling constant) compared to the previous instrument.



(a)



(b)

Figure 6: (a) Exploded schematic of a long stave (b) Example parasitic geometry for determining stave performance.

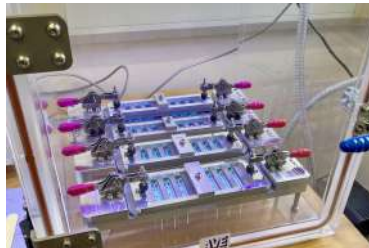
Super MuSR will facilitate these upgrades by using significantly faster readout electronics, and increasing the number of detector elements from 64 to 960. Two cylindrical barrels provide for optimal solid angle coverage, each of which are made of 16 staves made of 32 detector elements of varying length [16], each individual detector element is coupled to two Silicon Photomultipliers (SiPMs) that collect light produced by the scintillating tiles. SiPMs are solid state pixelated detectors, with similar gain to PMTs. They are extremely compact (1.3mm^2 active area) and magnetically immune [17], allowing the barrels to be placed inside the main MuSR magnet.

3.4.3 Scintillating Tile Design

To collect information about the positrons, a series of novel scintillating tiles have been designed and produced for Super MuSR. They come in 30mm, 43mm, 63mm, 105mm and 210mm sizes to conserve solid angle irrespective of distance from the sample. The tiles are made of Eljen 204 scintillator which has a long optical attenuation length and fast timing, ideal for the high rates of data collection Super MuSR aims to achieve [18]. The tile assembly process begins with inserting wavelength shifting fibre (WLSF) into the scintillator. WLSF is bent at 90 degrees in two places, allowing light produced by the scintillator to be directed towards the active area of the SiPMs. Cracks and damage to the bends can cause poor light output, so bends are done using a specialised jig, illustrated in Figure 7a, which is submerged in hot water and used to form the bends with minimal damage. The fibre and scintillator are held in place by a gluing/curing jig seen in Figure 7b, allowing the tiles to be in the correct position before attaching the WLSF to the scintillator with an optically transparent resin. Bubbles and imperfections in the coupling will reduce light output, so the solution is to degas the resin in a vacuum chamber. This is a three step process where first the container of resin is degassed, and then the resin is injected into the cavity, filled to the top, placed into a vacuum chamber and degassed twice more. Plastic studs are slotted onto the ends of the WLSF to provide mechanical strength to the WLSF, and to provide a method of alignment for the WLSF to the SiPMs. Once the resin is cured, the stud and WLSF will be lapped using high grit micro-fine silicon carbide sandpaper, such that they are the correct height and have a polished finish, this process is facilitated by a lapping jig shown in Figure 7c. The tile is fixed in place and polished down to the correct length, where the tiles will be ready for wrapping (Figure 5b shows an example 30mm tile before wrapping). The final stage of assembly is the wrapping stage, this involves wrapping the tile in Excellent Specular Reflector (ESR) to minimise light loss, it is kept tightly wrapped by thin layers of aluminium tape, providing maximum reflection from the ESR. Images of the tiles after wrapping can be found in the appendix (Section A.1).



(a) Fibre bending jig for the 105mm tiles. Straight fibres are lined up in the jig and bent to produce fibres with a 90° angle.



(b) Gluing and curing jigs for the 30mm, 43mm, and 63mm tiles, which are inserted in batches of 6 into the jig, filled with resin, degassed inside a vacuum chamber, and then placed into an oven to cure.



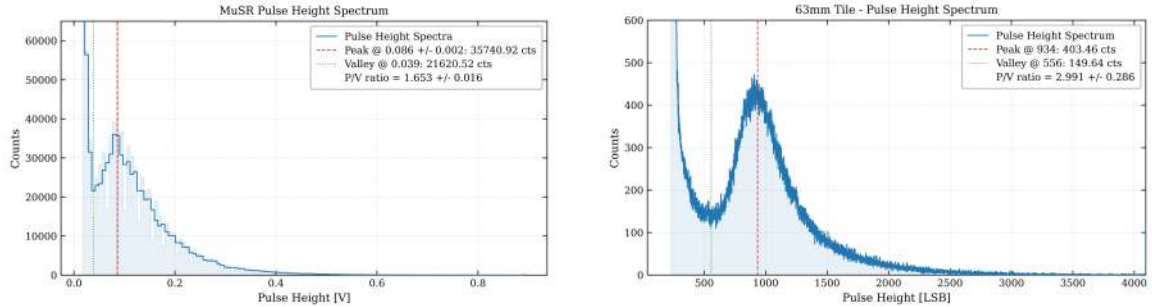
(c) Lapping jig for 30mm tiles, the tile is inserted into a stainless steel jig, fixed in place, and then lapped with micro-fine, silicon carbide sandpaper. This ensures studs are held in place, minimising damage to the WLSF, and also ensure the studs are lapped to the correct height.

Figure 7: (a) Fibre bending jig. (B) Gluing and Curing Jig inside Vacuum tank.(c) Lapping and Polishing Jig.

4 Experimental

4.1 Determining Acceptance Threshold

The acceptance threshold is a criteria that if exceeded, the tile will be approved for the final detector. Tiles that fall below this threshold are to be investigated for defects. In order to determine an acceptance threshold for the scintillating tiles, a series of benchmark tiles have to be determined, as testing all tiles on the beamline is not possible due to time constraints. A Pulse Height Spectra (PHS) is a histogram of different pulse amplitudes and their respective counts [14]. The Pulse Height Spectrum (PHS) peak-to-valley ratio (P/V) ratio was utilised as the Figure-of-Merit (FoM) for performance on the beamline. A better P/V ratio allows for more precise separation of the noise floor and low amplitude positron events. The overall criteria required for a given stave was an average P/V ratio greater than the current MuSR instrument. Figure 8 illustrates the difference in PHS between the current MuSR instrument and Super MuSR. PHS for all tile lengths can be found in the appendix (Section A.2). PHS from the old MuSR instrument were taken using an 8-bit oscilloscope, providing a P/V of 1.653 ± 0.016 , after being smoothed using a Savitzky-Golay filter [19]. For the Super MuSR tiles, multiple runs had to be conducted to properly align each Module with the beamline. Figure 6b illustrates that the detector stave is too long to fit all modules in the line of sight of the detector. In total 5 runs were needed to get the performance of all 32 tiles, which had an average P/V ratio of 2.715 ± 0.058 , a 64.5% increase compared to MuSR. This means that the ensemble of tiles can be used as benchmark tiles. As a result of the increased resolution of Super MuSR (12-bit compared to 8-bit), and the limited time per run, the counting statistics are worse, resulting in larger error in the Super MuSR data. On an individual tile basis, all tiles had a greater P/V ratio than MuSR except the 210mm tiles. P/V for 210mm tiles was 1.556 ± 0.133 , 6% worse than the current MuSR detector. Reasons for reduced performance involve the greater difficulty aligning the geometry. Optimal alignment of the 210mm tiles was not possible as incident positrons were not arriving perpendicular to the tile surface. Furthermore, these tiles are at least twice the length of other tiles, as solid angle was conserved at the expense of brightness. Longer tile lengths greatly increase the range of path lengths, and hence amount of attenuation present in these tiles. This broadens the peaks, making the P/V ratio smaller. Despite a lower P/V ratio, the larger horizontal separation will allow for effective separation between background noise and positron events. Furthermore the other tiles bring up the staves overall P/V ratio, which means these tiles are still suitable for the final detector.



(a) PHS taken of the MuSR instrument for a ZF with a silver sample. MuSR uses PMTs and so the Pulse Height is a function of Voltage.

(b) PHS of a 63mm tile taken parasitically on a Super MuSR detector Stave. Super MuSR uses SiPMs which have been converted from voltage to Least Significant Bit (LSB) which represents the smallest ΔV of the SiPM.

Figure 8: Comparison of PHS between MuSR and Super MuSR instruments.

One tile of each length was selected as a benchmark tile. The 30mm benchmark tiles had already been established in earlier work, (Dan Pooley Personal Communication, 2025), so although 30mm tiles were tested on the beamline they were not used as benchmark tiles. Newly produced tiles must be comparable to the light output of the benchmark tile. If a tile performs 2σ below the mean of the repeats of the benchmark tile, then it is investigated. If it exceeds this threshold then it is approved for the final detector. The pre-established threshold for the 30mm tiles is that all tiles must exceed the benchmark tiles brightness. Having a physical benchmark tile allows a consistent point of normalisation, therefore any changes that are made to the experimental setup can be corrected for. The weakness of using a single benchmark tile for each length is that damage to the benchmark tile prevents the ability to compare to previous batches of measurements taken with the tile before it was damaged. Tiles were still used as the benchmark as they are less sensitive to change compared to the experimental setup, PMT photocathode damage, damage to electronics and PMT gain drift were more common than damage to the scintillating tiles.

benchmark repeats vs single measurement?

4.2 Scintillating Tile Testing

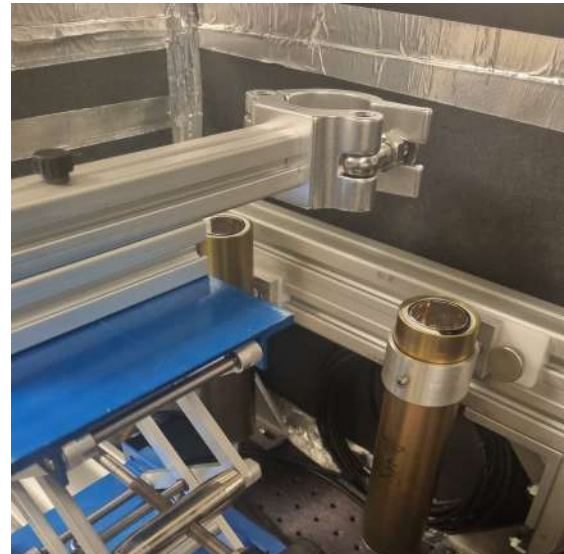
4.2.1 Experimental Setup

Now that a benchmark tile for each tile length has been established, light output independent of the beamline must be determined to ensure that tiles are suitable for the final detector. A collimated Strontium-90 (Sr^{90}) source is used to mimic positron emission, and is inserted into a bespoke rig which allows three dimensional movement. To account for the varying lengths of tile, a pair of PMTs are held in place on an optical rail so their separation can be varied. The tile testing box has aluminium tape over all gaps and an additional light tight blanket is placed over the top to ensure light tightness.

diagrams are far better than this



(a) A 210mm tile in place on two top of PMTs, each PMT is mounted on optical rail to facilitate measurement of all tile lengths.



(b) Scintillating tile testing rig with the 3D movement ^{90}Sr Holder.

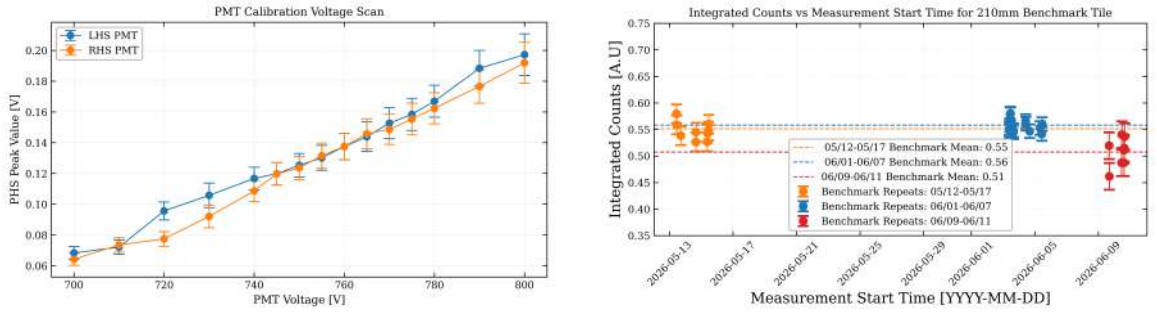
Figure 9: Images of the tiles testing rig, which was used to test all lengths of scintillating tiles.

4.2.2 Procedure

Each tile has a designated LHS, and RHS. Two runs are carried out per tile, one with the LHS stud coupled to the LHS PMT, and one with the tile reversed. This is done to account for any differences in the tile studs or PMTs themselves. Each tile is cleaned using a 70:30 mix of deionised water and isopropanol, as high concentrations of isopropanol have been shown to crack plastic such as the WLSF and studs of the tiles [20]. The tiles are then coupled using optical vacuum grease to the PMTs. After aligning the Sr^{90} , PHS are collected using a Picoscope, which gives a separate spectra for the LHS, RHS and total (LHS + RHS) PHS for any given tile. For 30mm tiles, the separation of the studs is too small to require both LHS and RHS PMTs, so both studs are coupled to a single PMT. On shorter timescales (within a day of measurements) the main source of error between runs is the optical coupling and alignment. This was confirmed by using a single 30mm tile. When taking a series of PHS without altering the optical coupling or alignment, maximum variation between runs was 1.2%. In contrast, removing, cleaning and redoing the optical grease and alignment each time, the variation between measurement increased to 6.2%. These two dominant sources of error can only degrade the light output, therefore the highest light output run for each tile was selected to get the best representation of tile performance. For the shorter tiles, the chosen FoM for light output is the PHS peak value, a greater PHS peak indicates a greater overall light output. However for the longer lengths of tiles (105mm and 210mm), PHS peaks are less prominent (see example PHS in Figure A3 in the appendix). The PHS peak finding code could not consistently find a good peak. Consequently the FoM was changed to integrated counts for these tile lengths. The peak finding algorithm utilises a Gaussian fitting to the peak after applying a Savitzky-Golay filter [19]. Integrated counts were calculated using the composite trapezoidal rule [21].

4.2.3 Calibration and Measurement Stability

The PMTs were gain matched by taking a PHS with the 30mm benchmark tile for a range of voltages on each PMT. The 30mm benchmark tile was coupled to each PMT, then PHS were taken at 10 minute intervals to strike the best balance between time efficiency and statistical quality of the measurements. As a result of the calibration both PMTs were set to a voltage of 760V. Throughout any given batch of measurements, the benchmark tile is measured regularly to provide



(a) Gain-matching scan of the scintillating tile testing setup. LHS and RHS PMTs were given voltages of 760V.

(b) A Stability plot with the 210mm benchmark tile showing variation in the experimental setup with respect to time.

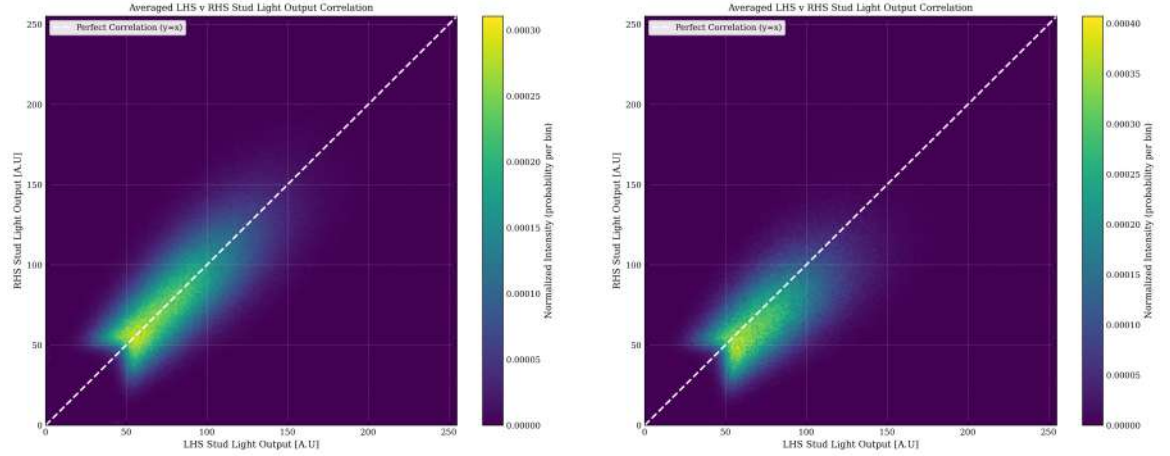
Figure 10: (a) Calibration of the tile testing rig. (b) Example stability plot.

a consistent point of comparison. Across a standard day of measurements two benchmark runs are done at the start and end of a day. Each batch is normalised to its benchmark measurements in order to account for error that occurs over a longer timescale such as PMT gain drift, HV power supply drift and ambient temperature changes. Figure 10b demonstrates an example of batches of

measurements, grouped with respect to time. The example shows up to 9.8% variation between batches of measurements. This source of error that occurs across longer timescales, and therefore is corrected for by normalising each measurement batch.

4.2.4 Stud Analysis

Each event measured on the Picoscope is the coincident of pulses from the LHS and RHS of the tile. For a performing tile it is expected that each stud has an equal light output. If one side of a tile is performing worse, it follows that one PHS has a lower light output than another. To detect this, a 2D pulse height correlation plot can be made to discern differences between the LHS and RHS of the tile. The pulse height of a given event is measured on both the LHS and RHS PMTs and then they are plotted against one another, forming a 2D heat-map. As there are two repeats conducted of each tile, the runs can be normalised and averaged to correct for difference between the PMTs. This results in an averaged PHS correlation plot which can be used to identify differences between the sides of the tile. There are two main modes of damage for each side of the tile, one is damage occurring during the WLSF bending process resulting in damage to the WLSF bends, decreasing the light output of a given side of the tile. Another cause of damage is the shearing forces exerted on the studs and WLSF throughout the lapping process.



(a) Averaged pulse correlation plot for a 210mm tile without a damaged stud.

(b) Averaged pulse correlation plot for a 210mm tile with a damaged stud.

Figure 11: Two tiles that have been analysed using the pulse height correlation code. (a) has a long and symmetrical distribution, compared to a shorter and slightly skewed tile (b). In tile (b) LHS stud is below the X-axis, meaning that stud has a greater light output, as the pulses are of greater intensity on the LHS PMT. For this particular tile when investigated the RHS stud was wobbly suggesting damage.

too subtle
to see in
this kind of
plot .

5 Results, Discussion and Analysis

5.1 30mm Tiles

As seen in Figure 12 the 30mm tiles are performing above the acceptance threshold, performing 23.7% better on average. Although only a single tile falls below the acceptance threshold, 5 tiles fall within two standard deviations of the benchmark mean and were therefore investigated and compared to the production teams notes on each tile.

ID	Resin	Production Notes
40	IN2	Divot both ends, dull crackly finish, bubble one corner, re-lapped
49	RAL71	Clear resin, bubble on one fibre leg, Some small scratches on tile, re-lapped
57	RAL71	Ultrasonic washed tile, clear resin, bubble in 1 stud, mark on tile, re-lapped
61	RAL71	Clear resin, bubble/air in one leg, re-lapped
64	RAL71	Clear resin, possible contamination in both legs, bubble in one stud, re-lapped
120*	RAL71	Tiny bubble in 1 stud and red spec in the other

Table 1: Production teams notes for poor performing 30mm tiles. (*) Note tile 120 has been included to show how the notes compare to a tile which meets the required light output. Tile IDs reset to 1 for each tile length discussed in this report, and represent the order in which the tiles were made.

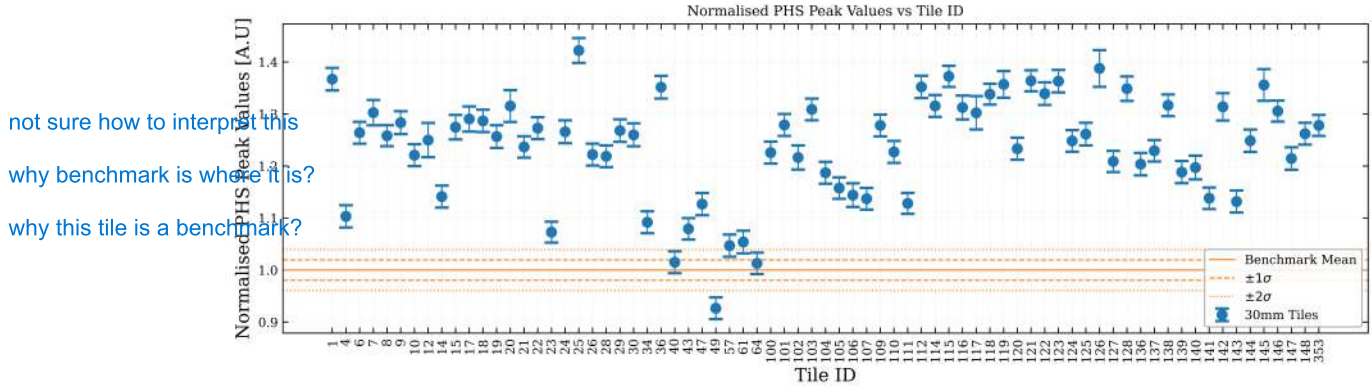
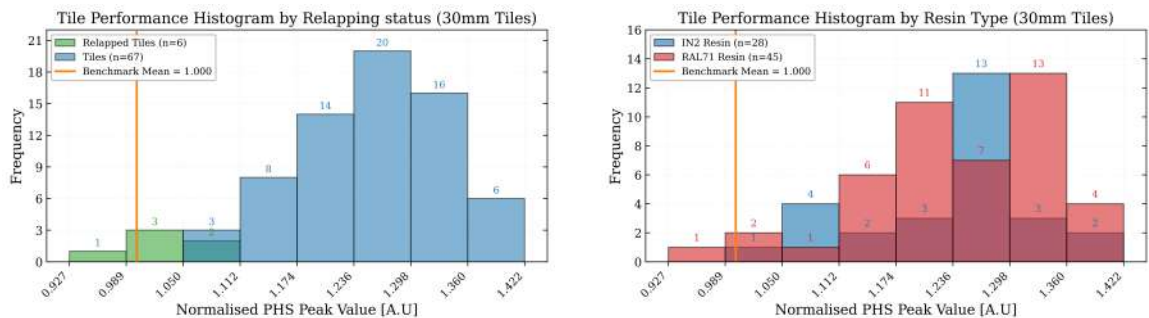


Figure 12: Scatter diagram of Pulse Height Spectra Peak Values vs Tile ID for the 30mm Tiles.

Two different clear resins were used throughout the tile production, IN2 Resin, which is an industry standard clear resin, with low absorption in the UV wavelength range [22], and an in house resin known as RAL71 resin, which has identical performance in terms of bulk transmittance [23], but has been noted as being harder to work with. As a result, initial tiles were made using IN2 resin as ease of handling was a key consideration for resin choice. Tiles 1 through 48 were made using IN2 resin, most of which performed well, despite a noticeable cloudy surface finish. Concern was raised that longer tiles will be affected more severely by the poor surface finish, due to increased area of cloudy resin on the tile. Table 1 outlines a "dull crackly finish" on the IN2 tile which is representative of all IN2 tiles, they were also described as "crusty" and "milky". As a result RAL71 resin was used from tile ID 48 onwards, despite being noted as harder to handle and use. 4/5 of the poorly performing tiles have RAL71 resin, which initially suggests that it has a lower light output, however after tile 100, not a single tile (all made with RAL71) has fallen within two standard deviations of the benchmark tile. Tiles were separated by resin and plotted on separate histograms (Figure 13b), which shows no clear difference in tile performance with respect to resin type. In order to determine if there is any statistically significant difference between tile performance for each resin type, the tiles were separated by resin type and a Student t-test [24] at a significance level of 0.05 was conducted. The resultant p-value was 0.756, showing that there is no statistically significant

difference in tile performance as a result of resin type, so RAL71 resin was still used to prevent the cloudy surface finish. All tiles have noticeable flaws such as bubbles, marks, scratches, and contamination as noted in the technicians notes, but these issues are found uniformly across all tiles, and so cannot be the sole cause of the investigated tiles failing. Another hypothesis for these poorly performing tiles that they were all re-lapped, these tiles were re-lapped as were required to initially have longer studs for some separate tests at the beginning of development. This means that the tiles had to be lapped for longer, and so were exposed to greater shearing forces compared to the other tiles, which had less stud length to lap down, causing an increased chance for cracks and damage in the WLSF and tile studs. There was a statistically significant result (t-test, $p=1.237 \times 10^{-9}$, significant), indicating this is a cause of reduced performance. Despite a small sample of re-lapped tiles ($n=6$), there was no evidence of outliers driving the low p-value. Therefore it can be concluded that re-lapping is the most probable cause of the decreased tile performance of the investigated tiles.

it would be good to show the defects



(a) Overlay of histograms with all 6 relapped tiles separated.

(b) Overlay of histograms of each resin type.

Figure 13: Two histograms of 30mm scintillating tiles. (a) Overlay of histograms with relapped tiles separated (b) overlay of histograms of each resin type.

The histogram in Figure 13a illustrates a negatively skewed Gaussian distribution, with the most tiles belonging to the 1.236-1.298 bin. The main population of tiles range from a performance of 1.112-1.360, which has 79.5% of the tiles inside this range. All 5 of the investigated tiles fall into the leftmost 3 bins. The asymmetric skew is expected as most tiles are manufactured near the maximum achievable light output for 30mm scintillating tiles, but damage and defects can drop tile performance, contributing to the tail of the distribution. Overall, the 30mm tiles have a failure rate of 1.4%, with tiles produced past tile ID 100 having zero tiles fall below the acceptance threshold.

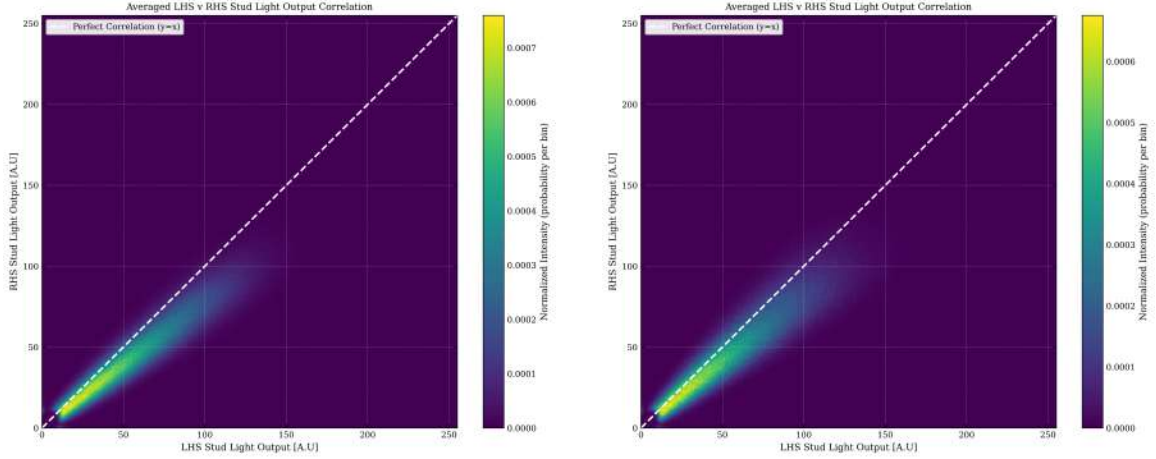
5.2 63mm Tiles

can you show them as image? it would be very good.

Each of the tiles have a series of small defects, mostly involving the stud of the tiles. When conducting 2D stud analysis, all investigated tiles except for tile ID 57, 32 and 36 were symmetric about the $y=x$ line. The two most asymmetric tiles were ID 57 and 32. The tile ID 36 2D stud analysis can be found in the appendix (Section A.6.1) with the rest of the investigated tiles. This noted asymmetry suggests that damage to one of the studs is reducing light output. The 2D analysis and technician notes have pointed towards damage to the studs in all investigated tiles. No statistically significant difference was found between damaged and non-damaged studs (t-test, $p=0.697$, not significant). Although stud damage likely contributes to some reduction in light output, no statistically significant difference was found. Therefore the stud damage alone cannot account for the investigated tiles falling below the acceptance threshold.

ok

The 63mm tiles were used to investigate a new, low viscosity, optically clear resin, RT152 [25]. This resin was one choice for the 105mm and 210mm tiles as RAL71 resin has to be cured at higher temperatures, causing the longer tiles to undergo warping and bending inside the curing and gluing



(a) Averaged pulse correlation plot for a 63mm ID 057, noted to have poor performance and a single loose stud.

(b) Averaged pulse correlation plot for a 63mm ID 32, noted to have poor performance. From the stud analysis this tile was found to have one tile performing worse than the other.

Figure 14: Two 63mm tiles observed to have asymmetrical light output. (a) Tile ID 057. (b) Tile ID 032.

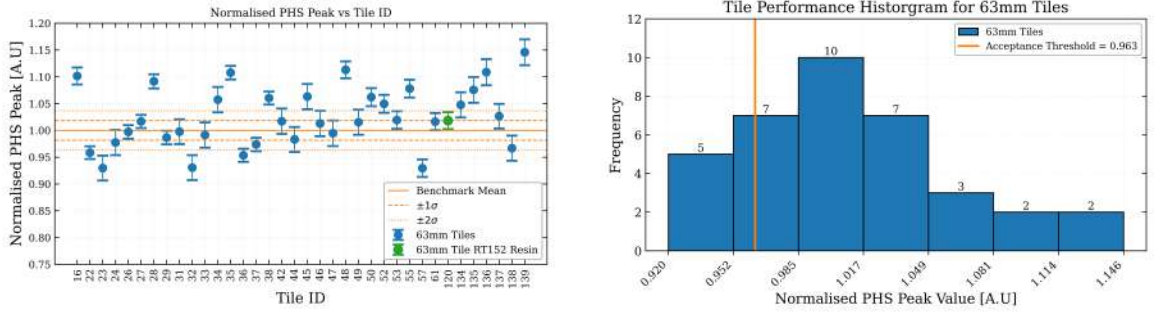
ID	Resin	Production Notes
22	RAL71	Resin topped up, tiny bubble in each stud, one low divot
23	RAL71	Resin topped up, bubble in each stud
32	RAL71	Large bubbles at bottom of each leg
36	RAL71	Tiny knick on tile
57	RAL71	1 loose stud
134*	RAL71	Semi chilled resin, scuff on top, tiny bubbles in 1 stud, hairs present

Table 2: Production teams notes for poor performing 63mm tiles. (*) Note tile 134 has been included to show how notes compare to a tile which meets the required light output.

jig. To prevent this issue RT152 was selected as a potential candidate. The curing time is 48hrs at 23°C , meaning the oven now simply maintains a controlled low humidity environment and does not cause any deformation in the tile. Tile 120 was made with RT152 resin in order to determine if the light output is suitable when using this resin. The tile falls above the acceptance threshold and therefore RT152 resin was used for the 105mm and 210mm tiles. Furthermore, two tiles that were verified on the beamline were also made using RT152 resin, confirming that it is a strong choice for resin. However this is a small sample size, so the longer tiles were closely monitored at the start of production to determine if this resin encounters any issues. The histogram shape is different to the 30mm tile performance histogram (Figure 13a), showing a Gaussian distribution with a positive skew.

This suggests that the 63mm tiles can still be improved, subsequently shifting the distribution

to the same negative skew pattern seen in Figure 13a. However, this could also be the result of the 63mm tiles not experiencing the same R&D phase the 30mm tiles had as the production process had been streamlined when production of all of the 63mm tiles began. For example the best resin had already been selected and the technicians already had lots of practice with the full tile production procedure. This means that the tiles are already centred around the maximum possible light output for the 63mm tiles. This is supported by the fact that the normalised standard deviation for the 63mm tiles is 0.055, compared to the 30mm with a standard deviation of 0.100. This means that the variation in tile performance is larger in the 30mm tiles, and that the 63mm tiles are being made more consistently, explaining why the negatively skewed Gaussian is not observed. Overall the tiles perform 1.5% better than the benchmark tiles, and therefore are comparable to the benchmark and suitable for the final detector. For the 5 tiles that were investigated, further measurements will be taken in order to determine if these tiles are suitable for the final detector, as although they fall below the acceptance threshold they are all within 10% of the benchmark tile. If a conservative approach is taken and all of these tiles are not used in the final detector, the failure rate for the 63mm tiles is 13.9%.



(a) Scatter diagram of Pulse Height Spectra Peak Values vs Tile ID for the 63mm Tiles.

(b) Histogram of all 63mm Tiles.

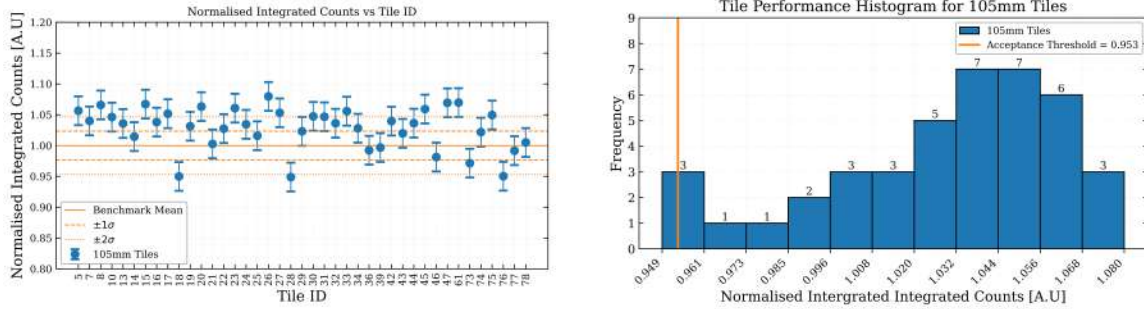
Figure 15: (a) Scatter diagram for the 63mm tiles. (b) Histogram for the 63mm tiles.

5.3 105mm Tiles

For the 105mm tiles, 41 tiles were tested and tile performance was measured using integrated counts. The tiles on average performed 2.9% better than the mean benchmark tile, therefore they are overall comparable to the selected benchmark tile. Once again the acceptance threshold was set 2 standard deviations below the benchmark repeats, 3/41 tiles were flagged for further investigation.

ID	Resin	Production Notes
18	RT152	Chilled Resin, Minor scuffs on tile, Small oxidation above 1 stud
28	RT152	Chilled Resin, Tiny hair in 1 stud, 1 Loose Stud
76	RT152	Semi Chilled Resin, Bubble in 1 stud
34*	RT152	Chilled Resin, Minor scratches on tile, Tiny hair and specks in 1 stud

Table 3: Production teams notes for poor performing 105mm tiles. (*) Note tile 34 has been included to show how notes compare to a tile which meets the required light output.



(a) Scatter diagram of Integrated counts vs Tile ID for the 105mm Tiles.

(b) Histogram of all 105mm Tiles.

Figure 16: (a) Scatter diagram for the 105mm tiles. (b) Histogram for the 105mm tiles.

Each of these tiles was noted to have an issue in a single stud, however so did the comparison tile that has performed above the acceptance threshold. No statistically significant difference between studs that were marked as damaged and those that were not was found (t-test, $p=0.810$, not significant). This means that stud damage cannot be solely responsible for the reduced light output of the investigated tiles. Figure A6 outlines the 2D stud analysis of all four of the tiles in Table 3. All three of the poorly performing tiles, and tile ID 34 exhibit asymmetry across one of the studs. Longer tiles experience greater levels of light attenuation and any small manufacturing defect has a longer length over which this defect can accumulate, resulting in asymmetry in each of the tile studs. Furthermore, the longer tiles may experience more trouble with optical coupling between the scintillator and WLSF due to the resin not flowing uniformly along the tile, resulting in non-uniform light output which would manifest in the form of asymmetrical light output, which was observed in the majority of 105mm tiles. SiPMs can be individually calibrated to account for a difference in light output between studs, so asymmetrical stud light output does not mean a tile cannot be used on the final detector. Another feature of note is "Chilled Resin" this refers to refrigerating the resin before using it, a process technicians integrated into the production process to improve the ease of handling of RT152 resin. Like the 30mm tiles, a negatively skewed Gaussian curve is obtained. The histogram is consistent with the pattern of most tiles being clustered near the maximum, and a long tail representing tiles with defects. For the 30mm tiles, this is due to the R&D process which is evidenced by the fact that there have not been any tiles past tile ID 100 that have failed, and tiles that fall below the acceptance threshold have done so in groups. For the case of the 105mm tiles the poorly performing tiles are spread uniformly throughout all tested tiles, suggesting that opposed to a systematic fault in production this is a feature of the randomness and difficulty of the production process. Overall the tiles were noted to have 7.3% failure rate, assuming all investigated tiles fail, with no clear indication of the cause of the drop-in performance. Current hypotheses attribute the reduction in light performance to a poor optical coupling between the WLSF and scintillator, caused by issues with the resin flowing non-uniformly across the length of the tile. It must also be noted that accounting for error these tiles may still have an acceptable degree of light output, and all 3 tiles fall within 5% of the benchmark mean. Therefore future work will involve re-testing these tiles in order to determine if they are suitable for the final detector.

you have also repeat errors. so each tile has some characteristics, but you measurement has about 15% uncertainty due to repeatability issues.

hard to follow in parts here

5.4 210mm Tiles

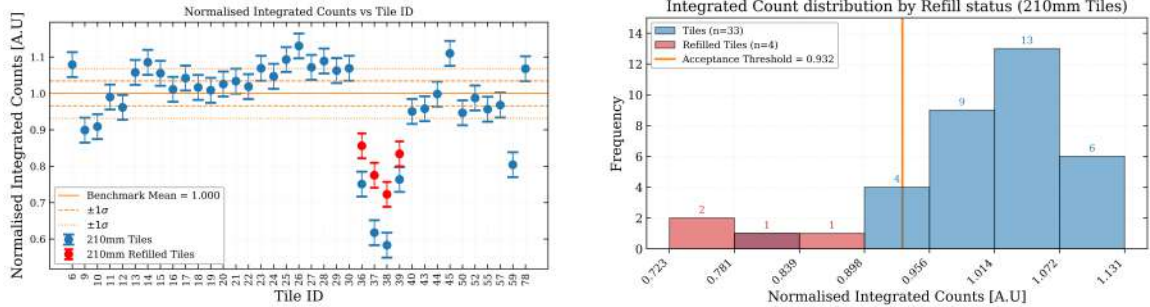
The 210mm tiles exhibited a severe drop in light output for a group of four tiles (36, 37, 38 and 39), outlined in red in Figure 17a. In total there were 37 tiles that were investigated, with an average of 1.9% worse than the benchmark tile repeats. When removing the four extremely poor performing tiles, the mean becomes a 1.7% improvement above the benchmark mean. Neither one of these percentages raise concern on their own, as this is a very small deviation away from the benchmark tile, however as this decrease comes largely from outliers, they were investigated alongside the other 3 tiles that fell below the acceptance threshold:

ID	Resin	Production Notes
9	RT152	Resin very messy - tiny hair in both studs - 1 low divot
10	RT152	Resin topped up, bubble in each stud
36	RT152	Resin a bit messy
36	RT152	Semi chilled resin, hair in resin
37	RT152	Semi chilled resin, small bubble in 1 stud, 1 high dome
38	RT152	Semi chilled resin
39	RT152	Semi chilled resin
40	RT152	Semi chilled resin
59	RT152	Semi chilled resin, Slightly high domes, bubble in 1 stud, tiny oxidation or contamination in the other, tiny Hair/debris evident
78*	RT152	Semi chilled resin, tiny hairs present real hairs or fibres

Table 4: Production teams notes for poor performing 210mm tiles. (*) Note tile 78 has been included to show how notes compare to a tile which meets the required light output.

No clear indication of faults in the four outlier tiles were found in the technicians notes, despite tile performance being 32.1% worse than the benchmark mean. They were also all from the same batch, so the decision to unwrap these tiles and inspect them was made. The inspection showed a very thin layer of resin along the top of the tile, illustrated in Figure A4. This is a mechanically suitable connection, as it holds the WLSF in place and is flush with the scintillator tile, allowing for the tile to be wrapped very easily and precisely. However, this appears to significantly reduce the light output, suggesting that conservative application of resin causes a much poorer optical coupling. Applying enough resin to the tiles such that it pools along the top of tiles guarantees the resin flows into the cavity between the WLSF and the scintillator along the whole length of the tile, forming a better optical collection and improving light output. In order to confirm this is the factor determining light output, the tiles were refilled and re-measured, as seen in Figure 17a. After refilling these tiles, they were re-measured, with an average improvement in light output of 17.4%. This strongly suggests that this is the reason for the failure of the 210mm tiles. However the refilled tiles did not recover all of the lost performance, and are still performing 20.3% worse than the benchmark mean. It is hypothesised that this is due to the interface between already cured and secondary applications of resin being less transparent than a single layer of resin. This could be due to grime, dust and

perhaps
visual
analysis is
not sufficient
to predict
performance



(a) Scatter diagram of Integrated counts vs Tile ID for the 210mm Tiles. Red represents the refilled tiles.

(b) Histogram of all 210m tiles separated by if they have been refilled or not.

Figure 17: (a) Scatter diagram for the 210mm tiles. (b) Histogram for the 210mm tiles.

other contaminants forming on the original application of resin. Furthermore this top up will not help the gaps between the WLSF and scintillator where the resin has already cured over the top, meaning some gaps cannot be improved by refilling the tiles. After this investigation all 210mm tiles were made with a larger amount of resin to ensure a strong optical coupling. Tile ID 078 is the first to have this larger amount of resin applied to it and exhibits a light output that has returned to above the benchmark tile, suggesting that the production team is now producing tiles that are now exceeding the acceptance threshold. Aside from the refilled tiles, tile IDs 9, 10 and 59 were noted to have damage to one or both of the studs, the 2D stud analysis (Figure A7) found that there was asymmetrical light output in tile IDs 10 and 78. No clear fault was found in Tile 78, and it is performing above the acceptance threshold and tile 10 had very loose stud. The histogram illustrates a negatively skewed distribution of tiles, with the leftmost tail being made of the four refilled tiles and tile ID 59. This represents the expected pattern of tiles being produced near the maximum possible light output with defects causing the left hand side tail of the distribution of the histogram. Unlike the 105mm tiles, the tiles that fall below the acceptance threshold do so more significantly compared to the 105mm tiles, of which all tiles were performing within 5% of the benchmark mean. Comparing this to the 210mm tiles, two tiles are within 10%, and one is within 20% of the benchmark mean (excluding the refilled tiles). This suggests that 210mm tiles are more greatly affected by defects compared to shorter tiles. Overall 18.9% of the 210mm tiles fell below the acceptance threshold, significantly higher than any other length of tile seen so far. However a more representative estimate of the failure rate going forward would be to exclude the four refilled tiles giving an overall failure rate of 8.8%.

6 Conclusions and future work

The failure rate and hypothesised causes for the failure have been outlined below in Table 5. Tiles will continue to be monitored until the first full barrel tests in February 2027 to ensure quality of all tiles. Due to many tiles showing no clear indication of faults in the technicians workbook, and refilling caused a significant increase in performance for 210mm tiles, the main mode of failure is hypothesised to be because of poor optical coupling between the WLSF and the scintillator as a result of insufficient resin being applied. Overall the production process and quality assurance methods are sound, because the cause for large groups of clustered failures such as re-lapping and resin application have been identified and fixed. Future work will involve determining an absolute rejection criteria, since acceptance threshold measures performance relative to a benchmark tile as opposed to absolute brightness.

very basic level of analysis.

there is no attempt to draw any other sources from literature or technical documentation on similar problems/solutions

The is no attempt on modelling coupling from WLSF to resin, or some micro-structure analysis, or other ways to get more detailed understanding of the interface between fibre and resin and optical/mechanical properties. This is a physics project, so one would expect more detailed insight, rather than stating observations.

Tile Length	% Below Acceptance Threshold	Hypothesised Cause
30mm	1.4%	Re-lapping and double handling (0 failures since tile ID 100)
63mm	13.9%	No confirmed cause, tiles to be investigated further
105mm	7.3%	No confirmed cause, hypothesised poor optical coupling between WLSF and scintillator
210mm	8.8% (18.9% including refilled tiles)	Poor optical coupling due to insufficient resin application
Total	8.6%	Double handling and WLSF-scintillator coupling are overall the most detrimental factors to tile performance

Table 5: Tile failure rates and hypothesised failure modes for each tile length.

what beamline testing would do differently?

Further beamline tests on investigated tiles will be conducted in order to determine if they are suitable for the final detector. A method of quantifying the asymmetry between studs is in development, and the 43mm tiles are the current priority for being tested on the PMTs, as they have had the fewest overall tiles tested. Full stave tests (32 tiles) on the beamline will be conducted to verify staves for the final detector. Furthermore development of a flood field testing rig to test entire modules (8 tiles) simultaneously is to be designed for the purpose of verifying modules independently of the beam, and for diagnosis of faulty modules after the first full barrel tests in February 2027. Overall this work has and will continue to contribute to the progress of the Super MuSR, which is a crucial part of ISIS endeavour programme, and will drive research and innovation for years to come.

there is little beyond simple observation and guessing. lots of testing material, but analysis is very rudimentary

A Appendix

A.1 Images of Completed Scintillating Tiles

at least one should have been in main text.

Below is a set of images showing the tiles after they have been fully wrapped. The Tiles go from longest on the left (210mm), to the shortest on the right (30mm) :



(a) Diagonal view of scintillating tiles.



(b) Plan view of all scintillating tiles.



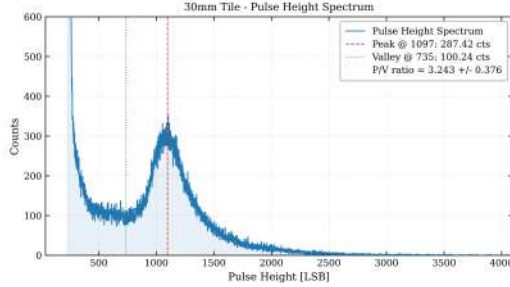
(c) Side-on plan view of scintillating tiles.

why thickness is different?

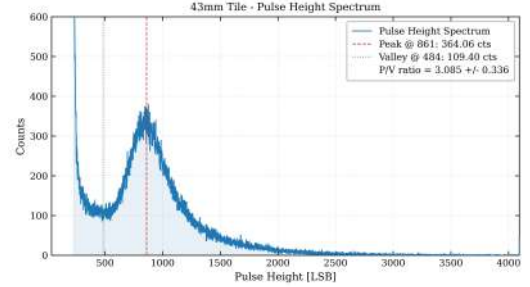
Figure A1: Views of the scintillating once fully assembled.

A.2 Beamline PHS for each Tile Length

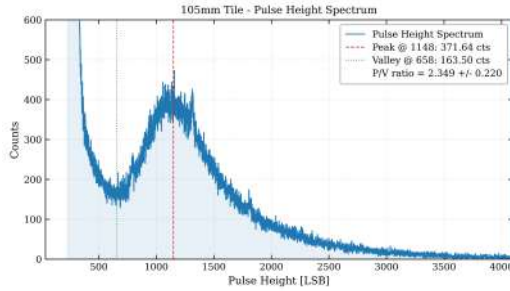
Below are example PHS for tiles of all lengths that were obtained from parasitic measurements on the beamline (excluding the 63mm tile which can be found in 8b). Although the 30mm tile benchmark tile was not determined using this method, and the 43mm tiles were not investigated within this report, they are still included for completeness:



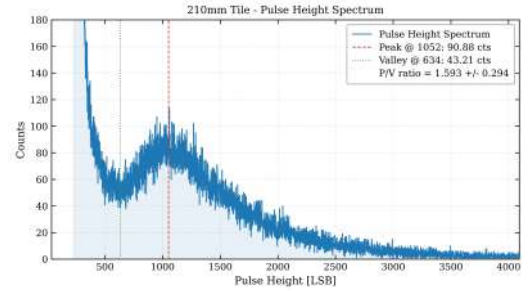
(a) PHS for a 30mm Tile.



(b) PHS for 43mm Tile.



(c) PHS for a 105mm Tile.

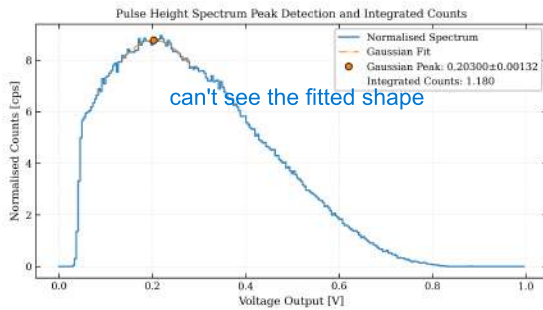


(d) PHS for a 210mm Tile.

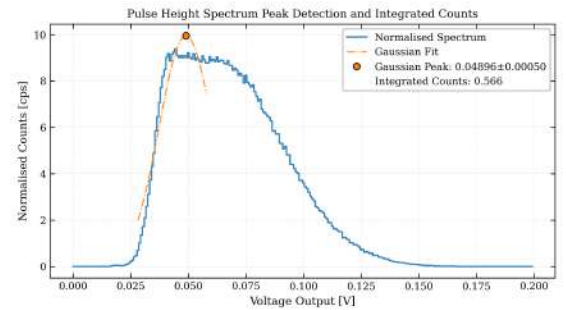
Figure A2: PHS for all tile lengths on Super MuSR

A.3 Example PHS

Below is an example of two PHS from a 30mm tile, and a 210mm tile. This figure illustrates why the longer tiles had to have integrated counts used as a FoM, as the Gaussian peak fitting is poor.



(a) Example PHS for a 30mm tile, with a Gaussian fitting.



(b) Example PHS for a 210mm tile with attempted Gaussian fitting.

Figure A3: Example PHS for (a) a 30mm Tile. (b) a 210mm tile.

A.4 Error Propagation

A.4.1 Beamline Peak to Valley

The peak to valley ratio is computed as:

$$P/V = \frac{N_P}{N_V} \quad (8)$$

Where N_p is the number of counts at the peak, and N_V is the number of counts at the valley. Assuming Poisson counting statistics, the error on the number of counts for single measurement is $\sigma = \sqrt{N}$, where N is the number of counts. The error on the P/V is:

$$\sigma_{P/V} = \frac{N_P}{N_V} \sqrt{\frac{1}{N_P} + \frac{1}{N_V}} \quad (9)$$

This error does not account for the error in the peak or valley fitting, only the statistical uncertainty of the counts [14].

A.4.2 Tile Testing Rig

To estimate the error experimental setup (σ_{exp}), repeats were conducted of the benchmark tile, subsequently determining the uncertainty that is a result of the experimental system, such as variation in optical coupling, HV gain drift and changes in the PMTs. The standard deviation of N measurements $x_1, x_2, \dots, x_N$ is given by [26]:

$$\sigma = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2} \quad (10)$$

The standard error on the mean is given by [26]:

$$SEM = \frac{\sigma}{\sqrt{N}} \quad (11)$$

Therefore, for all of the tiles measured where the highest performing run was selected, $N=1$ and hence the experimental error in the system is approximated by the standard deviation of the system. Where possible this error is added in quadrature with other sources of error, such as the peak finding code.

A.4.3 Scintillating tiles PHS Peak Value

There are two sources of error on the PHS peak value, the error in the fitting, and the inherent randomness of the experimental setup. The error in the Gaussian fitting is calculated using the built in error function on `curve_fit`. The uncertainty on the peak position (σ_{pp}) using `curve_fit` is given by:

$$\sigma_{pp} = \sqrt{\text{pcov}[1,1]} \quad (12)$$

This is added in quadrature with the error on the experimental system (σ_{exp}), Therefore the total error for tiles measured using the PHS peak value is :

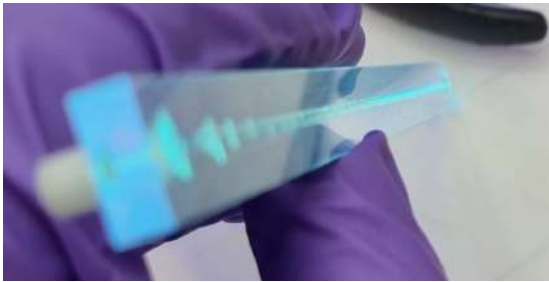
$$\sigma_{tot} = \sqrt{(\sigma_{pp})^2 + (\sigma_{exp})^2} \quad (13)$$

A.4.4 Scintillating tiles Integrated Counts

As `np.trapz` has no built in error function, and estimating error involves knowing the function of the PHS distribution [27], which was not determined in this work. Therefore error is estimated using only the error of the experimental setup. Future work will involve developing a method of estimating error on the integrated counts.

A.5 Application of Resin on 210mm tiles

Below in Figure A4, two pictures of tiles with and without a suitable amount of resin applied to them.



(a) An example of a "thin" bead of resin along the top of a 210mm tiles
 not sure what significance this has?



(b) An example of a more suitable bead of resin, the pooling effect observed has been noted to provide better optical coupling

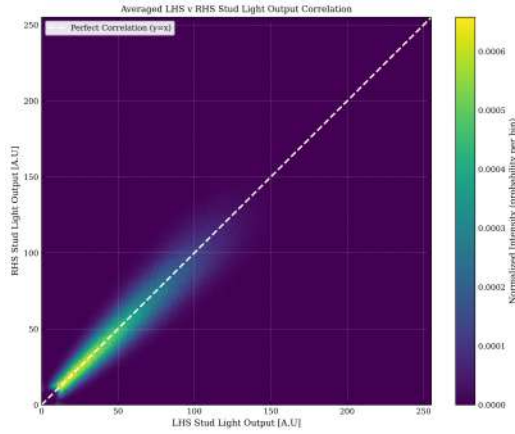
don't understand what I'm looking at here

Figure A4: (a) A thin bead of resin on a 210mm tile (b) A thicker and pooling bead of resin on a 210mm tiles

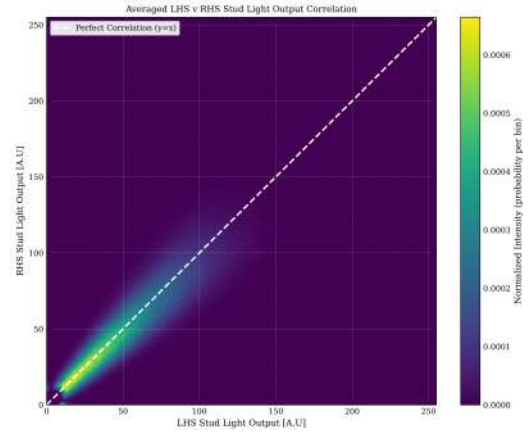
A.6 Pulse Height Correlation Plots

A.6.1 63mm Tiles

Below are the remainder of the Pulse Height correlation plots not found in Section 5.2:



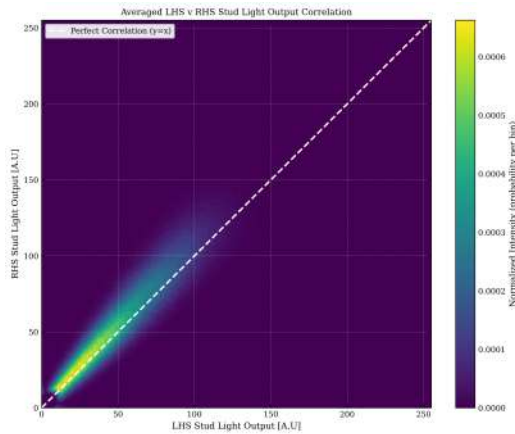
(a) Tile ID 18.



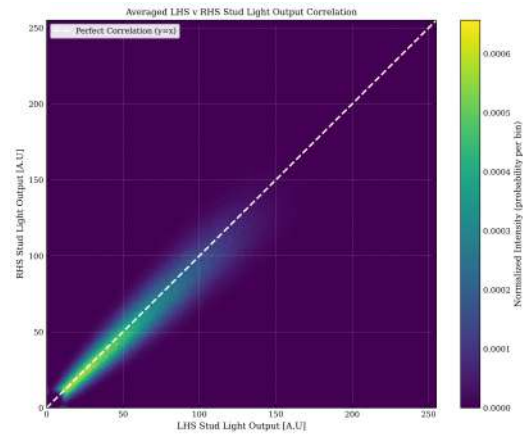
(b) Tile ID 23.

you need to display
asymmetry in a
different way

plot total signal
above and below
 $y=x$



(c) Tile ID 36.

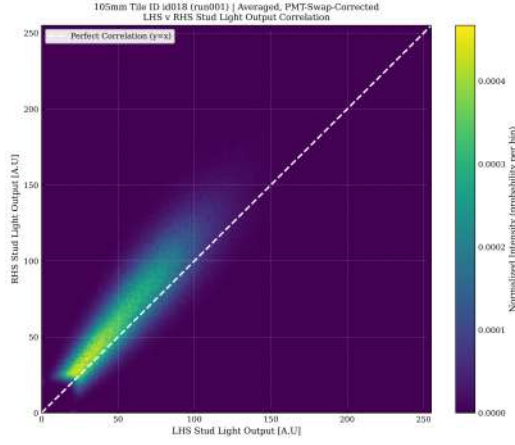


(d) Tile ID 134.

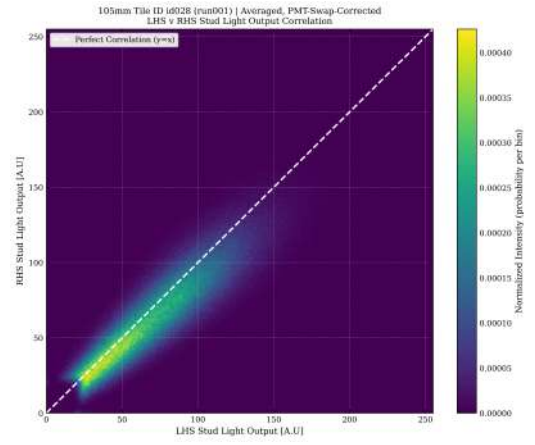
Figure A5: 2D stud analysis for all of the investigated 63mm tiles, and a comparison tile ID 134.

A.6.2 105mm Tiles

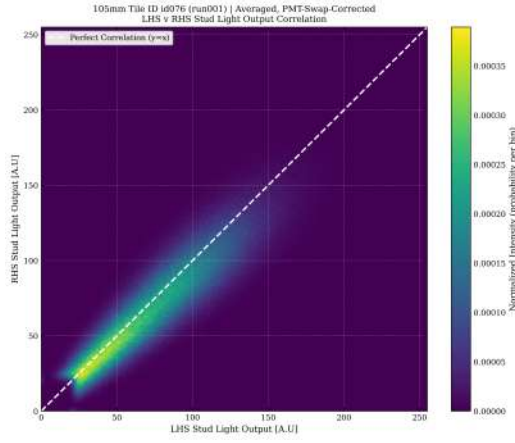
The tile 2D plots for the 105mm tiles can be found in Figure A6:



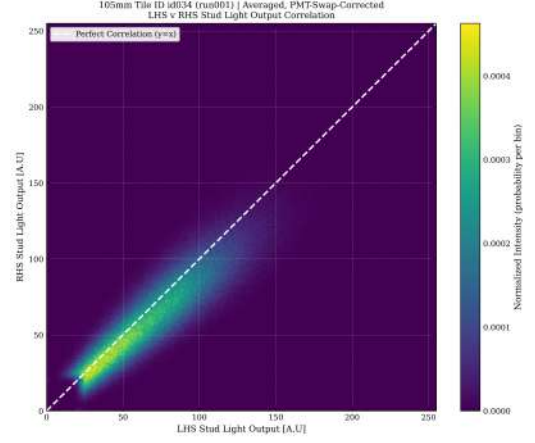
(a) Tile ID 18.



(b) Tile ID 28.



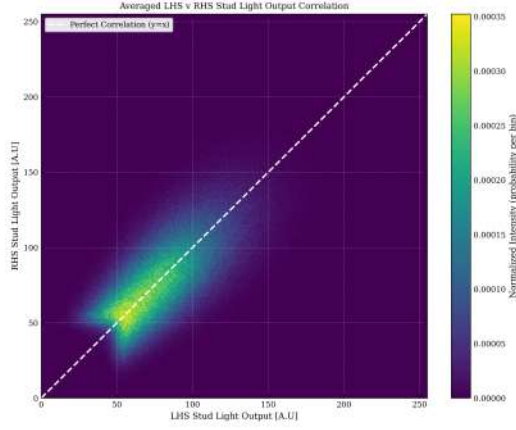
(c) Tile ID 76.



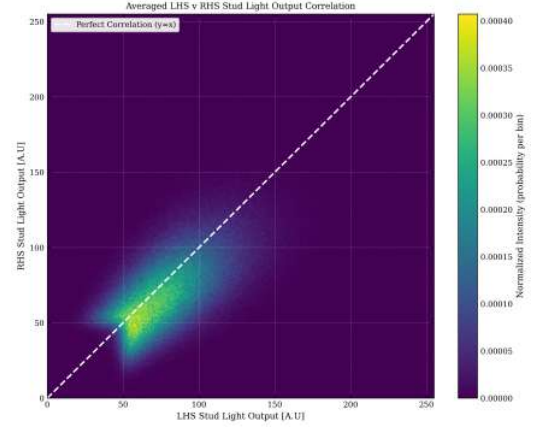
(d) Tile ID 34.

Figure A6: 2D stud analysis for all of the investigated tiles, plus a comparison tile.

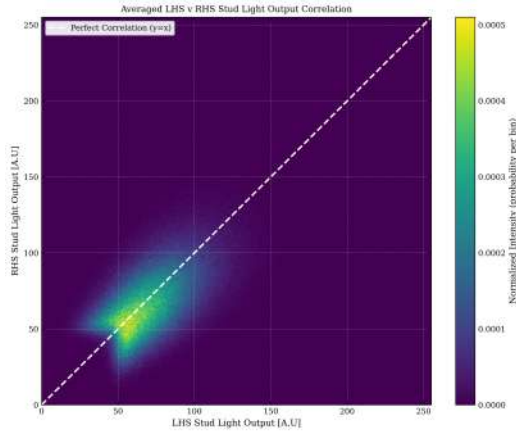
A.6.3 210mm Tiles



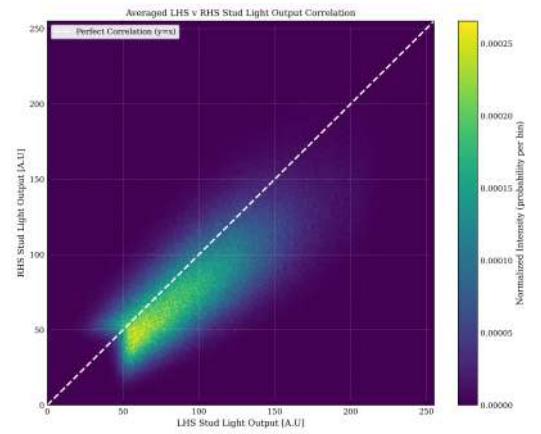
(a) Tile ID 9.



(b) Tile ID 10.



(c) Tile ID 59.



(d) Tile ID 78.

Figure A7: 2D stud analysis for all of the investigated tiles that were not refilled, but still fell below the acceptance threshold. Plus a comparison tile.

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